

06 Vector Space

6.3 Basis

A Basis Is Never Unique — from Standard Basis to Non-orthogonal Basis, via RGB/CMY Color Space



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

A basis is never unique: the same vector has different coordinates under different bases

2

Standard basis: the most "natural" unit-square grid, e.g. $\{\mathbf{e}_1, \mathbf{e}_2\}$

3

Orthonormal basis: a rotated/mirrored version of the standard basis — still pairwise orthogonal, unit length

4

Orthogonal basis: basis vectors are pairwise orthogonal, but need not be unit vectors

5

Non-orthogonal basis: basis vectors need not be orthogonal, but must still be linearly independent

6

Use the inverse matrix $\mathbf{A}^{-1}$ to find a vector's coordinates under a new basis

7

Change of basis: converting the same vector's coordinates between different bases

From Standard Basis to Non-orthogonal Basis, Constraint by Constraint

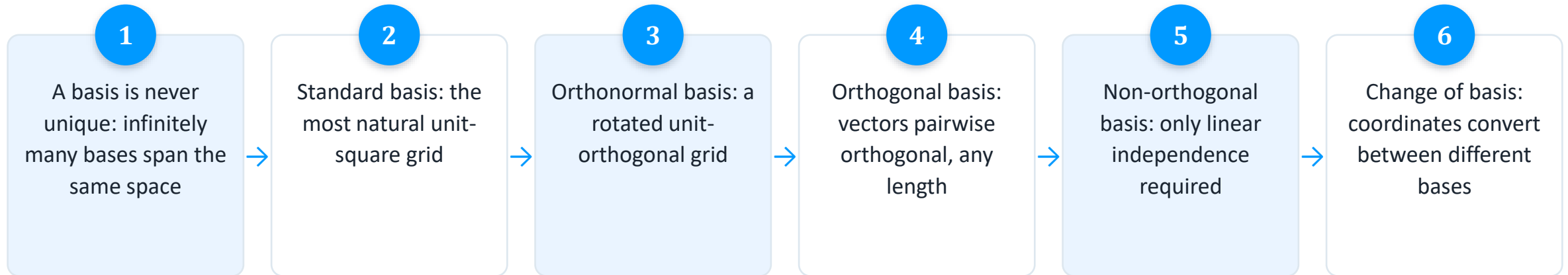
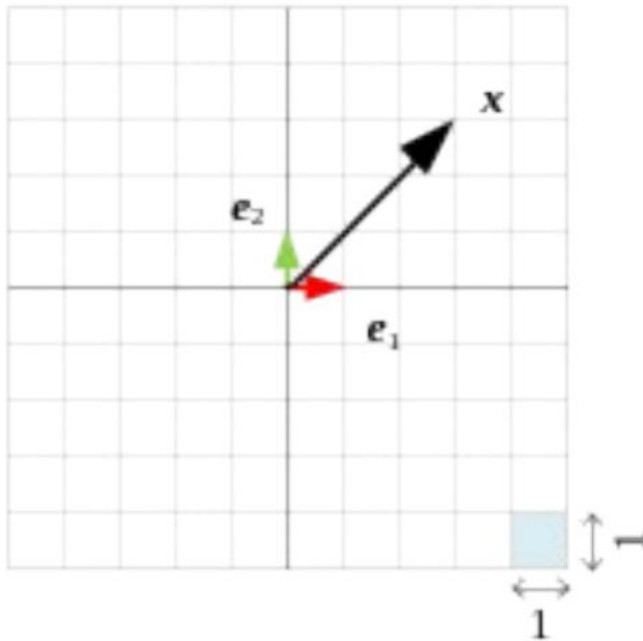


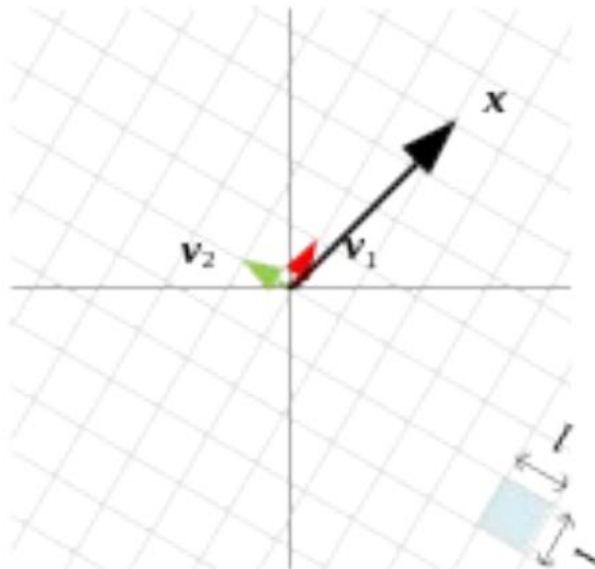
Figure 1 · Vector x in Three Different Orthogonal Coordinate Systems

- In (a)(b)(c), $\{e_1, e_2\}$, $\{v_1, v_2\}$, $\{w_1, w_2\}$ are all bases of the plane, each giving x a unique coordinate

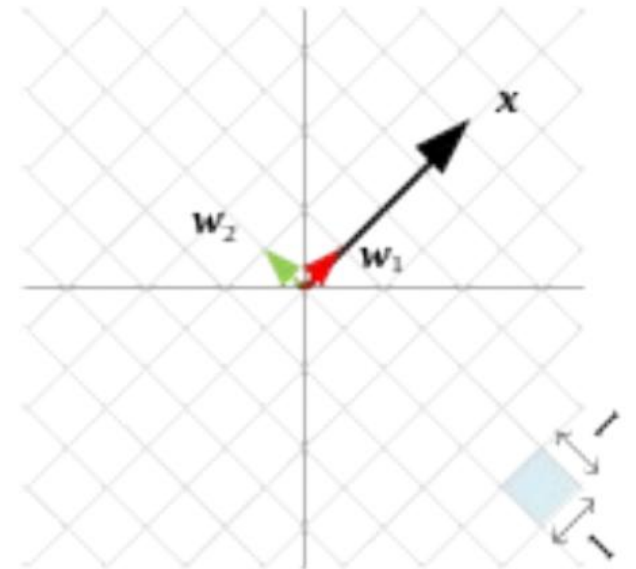
(a)



(b)



(c)



The Same Space Has Infinitely Many Bases

$$\mathbb{R}^2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2) = \text{span}(\mathbf{v}_1, \mathbf{v}_2) = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$$

- Every basis vector here is a unit vector, and within each pair they are orthogonal — such a basis is an orthogonal basis
- If every vector in an orthogonal basis also has unit length, it is called an orthonormal basis

Figure 2 · Basis Types: Non-orthogonal → Orthogonal → Orthonormal → Standard

- Constraints tighten step by step, from a non-orthogonal basis to the most "natural" standard basis

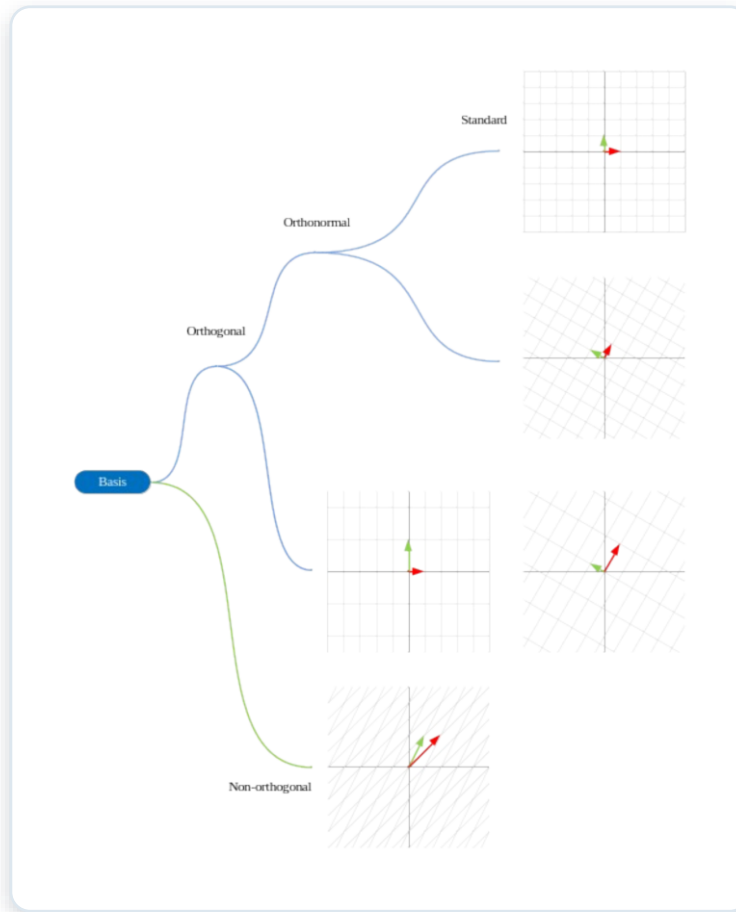
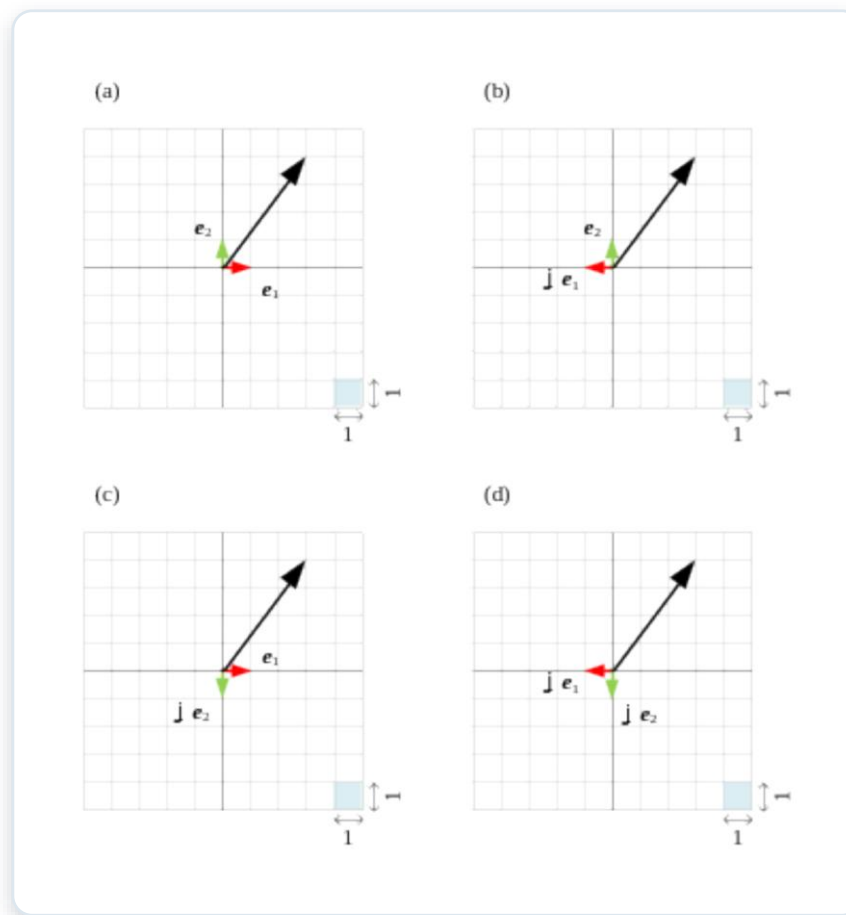


Figure 3 · A Few Unit-Square Grids

- Only (a)'s $\{e_1, e_2\}$ is the standard basis; (b)(c)(d) look "awkward" — they are orthonormal bases



COORDINATES OF $\mathbf{a} = [3, 4]^T$

Using the Inverse Matrix to Find Coordinates Under a New Basis

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{a}$$

- Vector $\mathbf{a}$'s coordinate $\mathbf{x}$ under basis $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2]$ satisfies $\mathbf{Ax} = \mathbf{a}$, so $\mathbf{x} = \mathbf{A}^{-1}\mathbf{a}$
- The standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ corresponds to the identity matrix $\mathbf{I}$, whose inverse is itself — so $\mathbf{a}$'s coordinate is just $(3, 4)$

Vector Coordinates Under a New Basis

Use `numpy.linalg.inv()` for the inverse matrix, then `@` to convert coordinates

- `np.array()` defines vector $\mathbf{a}$ and basis matrix $\mathbf{A}$ (swap the comment to try a different basis)
- `A_inv = np.linalg.inv(A)` computes the inverse of $\mathbf{A}$
- `A_inv @ a` gives vector $\mathbf{a}$'s coordinate under the new basis

```
## 初始化
import numpy as np

## 定义向量
a = np.array([[3],
              [4]])

## 定义不同基底
# 规范正交基
A = np.array([[ -1,  0],
              [ 0, -1]])

# 规范正交基
# A = np.array([[3/5, -4/5],
#              [4/5, 3/5]])

# 正交基
# A = np.array([[ -4/5, -3/2],
#              [3/5, -2]])

# 非正交基
# A = np.array([[2, 1],
#              [1, 3]])

## 计算逆矩阵
A_inv = np.linalg.inv(A)
A_inv

## 计算坐标
A_inv @ a
```

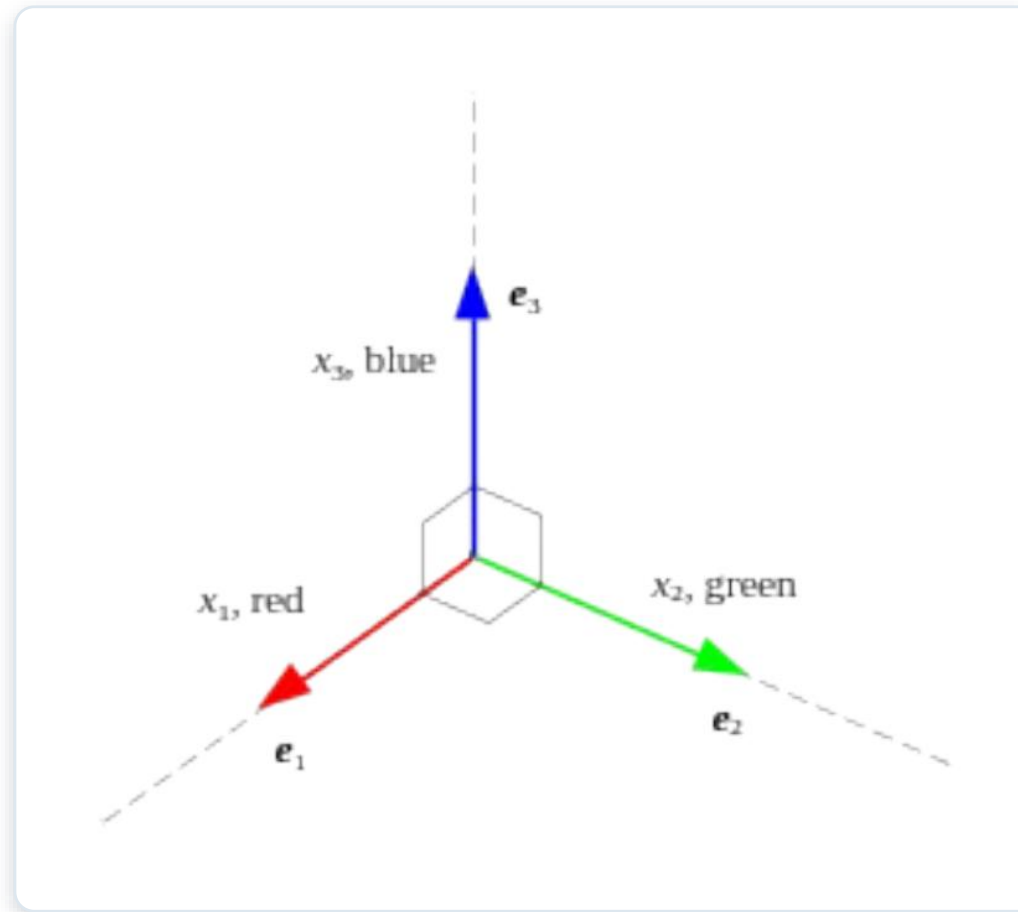
The Standard Basis $\{e_1, e_2, e_3\}$ of the Three-Primary-Color Space

$$e_1 \cdot e_2 = 0, e_1 \cdot e_3 = 0, e_2 \cdot e_3 = 0$$

- e_1, e_2, e_3 represent red, green, blue respectively — pairwise orthogonal and all unit vectors
- Any color = $x_1 e_1 + x_2 e_2 + x_3 e_3$, where x_1, x_2, x_3 range over $[0, 1]$

Figure 4 · The Standard Basis of RGB Color Space

- e_1, e_2, e_3 are pairwise orthogonal and unit length — the standard basis of the color space V



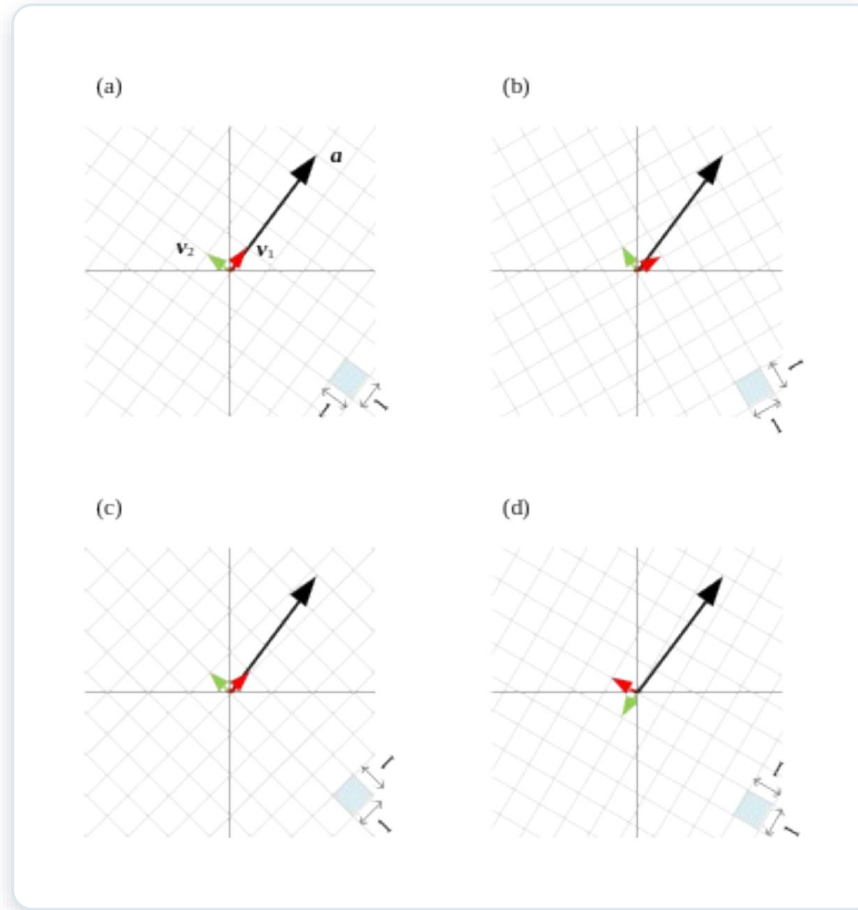
An Orthonormal Basis Corresponds to an Orthogonal Matrix

$$\mathbf{V}\mathbf{V}^T = \mathbf{V}^T\mathbf{V} = \mathbf{I}$$

- An orthonormal basis is a set of pairwise orthogonal unit vectors — it need not be as "natural" as the standard basis
- A matrix $\mathbf{V}$ satisfying $\mathbf{V}\mathbf{V}^T = \mathbf{V}^T\mathbf{V} = \mathbf{I}$ is called an orthogonal matrix

Figure 5 · A Few Rotated Unit-Square Grids

- Vector $\mathbf{a} = [3, 4]^T$ has coordinate $(5, 0)$ under the orthonormal basis in (a)



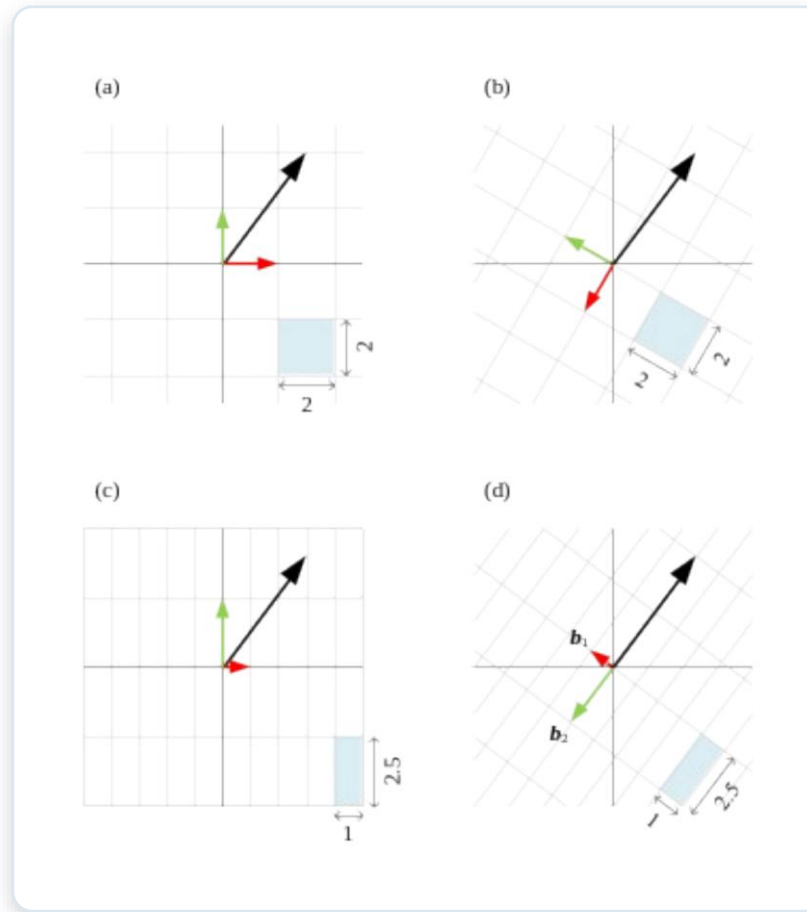
Drop "Unit Length", Keep "Orthogonal"

$$\mathbf{b}_1 \cdot \mathbf{b}_2 = 0$$

- An orthogonal basis requires basis vectors to be pairwise orthogonal, but not of unit length
- The grid can be a square, rotated square, rectangle, or rotated rectangle

Figure 6 · A Few Orthogonal Bases

- In (d), $\mathbf{b}_1$ and $\mathbf{b}_2$ have zero inner product — orthogonal — but $\mathbf{b}_2$ is not a unit vector



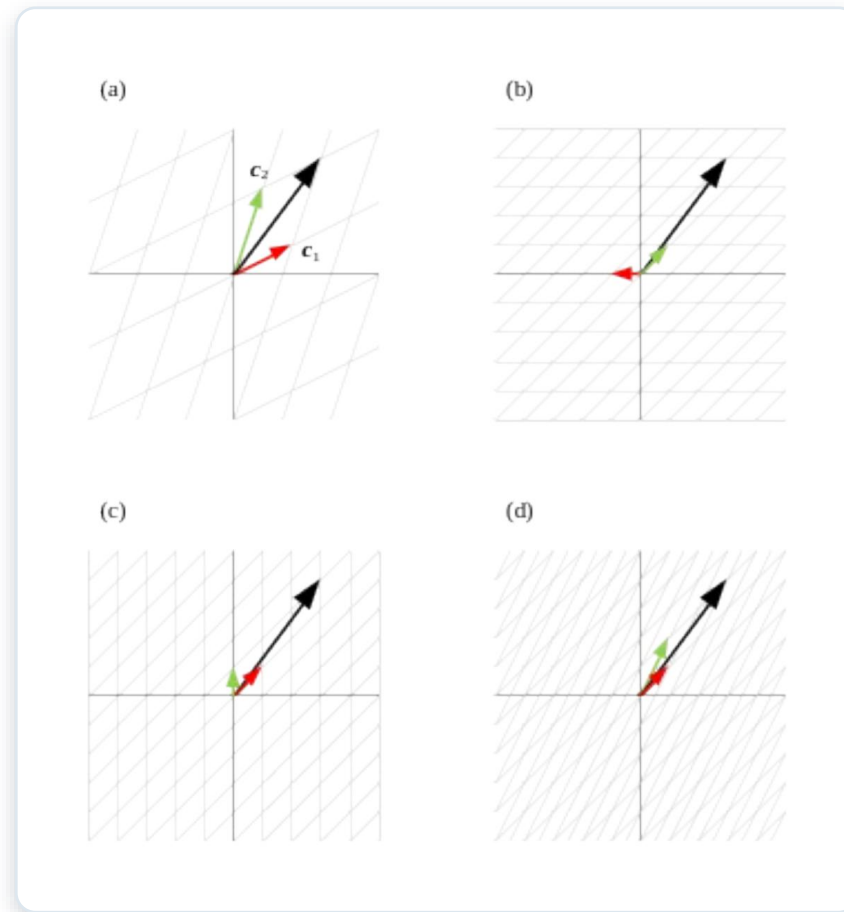
Relax "Pairwise Orthogonal", Keep Linear Independence

$$\mathbf{c}_1 \cdot \mathbf{c}_2 \neq 0$$

- A non-orthogonal basis drops the orthogonality constraint, but its vectors must still be linearly independent
- The grid is made of parallel, evenly-spaced parallelograms through the origin

Figure 7 · A Few Non-orthogonal Bases

- In (a), $\mathbf{c}_1$ and $\mathbf{c}_2$ have nonzero inner product; vector $\mathbf{a}$'s coordinate under this basis is (1, 1)



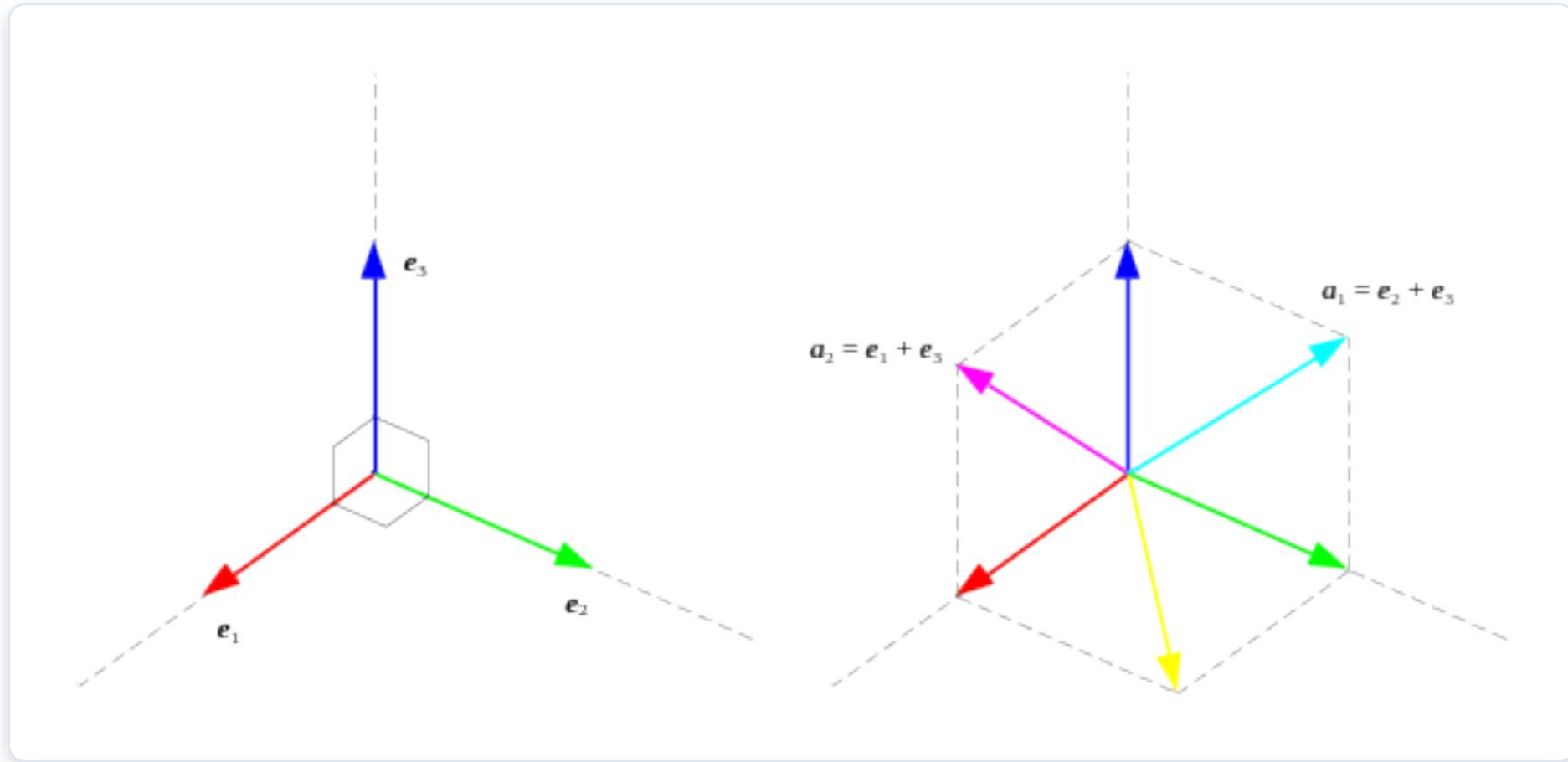
The CMY Non-orthogonal Basis $\{a_1, a_2, a_3\}$

$$a_1 = e_2 + e_3, a_2 = e_1 + e_3, a_3 = e_1 + e_2$$

- a_1 (cyan), a_2 (magenta), a_3 (yellow) each pair at a 60° angle — linearly independent but not orthogonal
- The three color channels of the CMYK printing model are exactly built on the non-orthogonal basis $\{a_1, a_2, a_3\}$

Figure 8 · From the Standard Basis to a Non-orthogonal Basis

- a_1, a_2, a_3 are linear combinations of e_2 & e_3 , e_1 & e_3 , and e_1 & e_2 respectively



Converting the Same Vector's Coordinates Between Bases

$$\mathbf{y} = \mathbf{W}^{-1}\mathbf{U}\mathbf{x}, \mathbf{x} = \mathbf{U}^{-1}\mathbf{W}\mathbf{y}$$

- Vector $\mathbf{v}$'s coordinates under bases $\mathbf{U}$, $\mathbf{W}$ are $\mathbf{x}$, $\mathbf{y}$ respectively, satisfying $\mathbf{U}\mathbf{x} = \mathbf{W}\mathbf{y} = \mathbf{v}$
- Combining the two equations lets you compute one basis's coordinates from the other's

Figure 9 · Change of Basis

- The same vector has coordinate (x_1, x_2) under basis $\{u_1, u_2\}$ and (y_1, y_2) under basis $\{w_1, w_2\}$

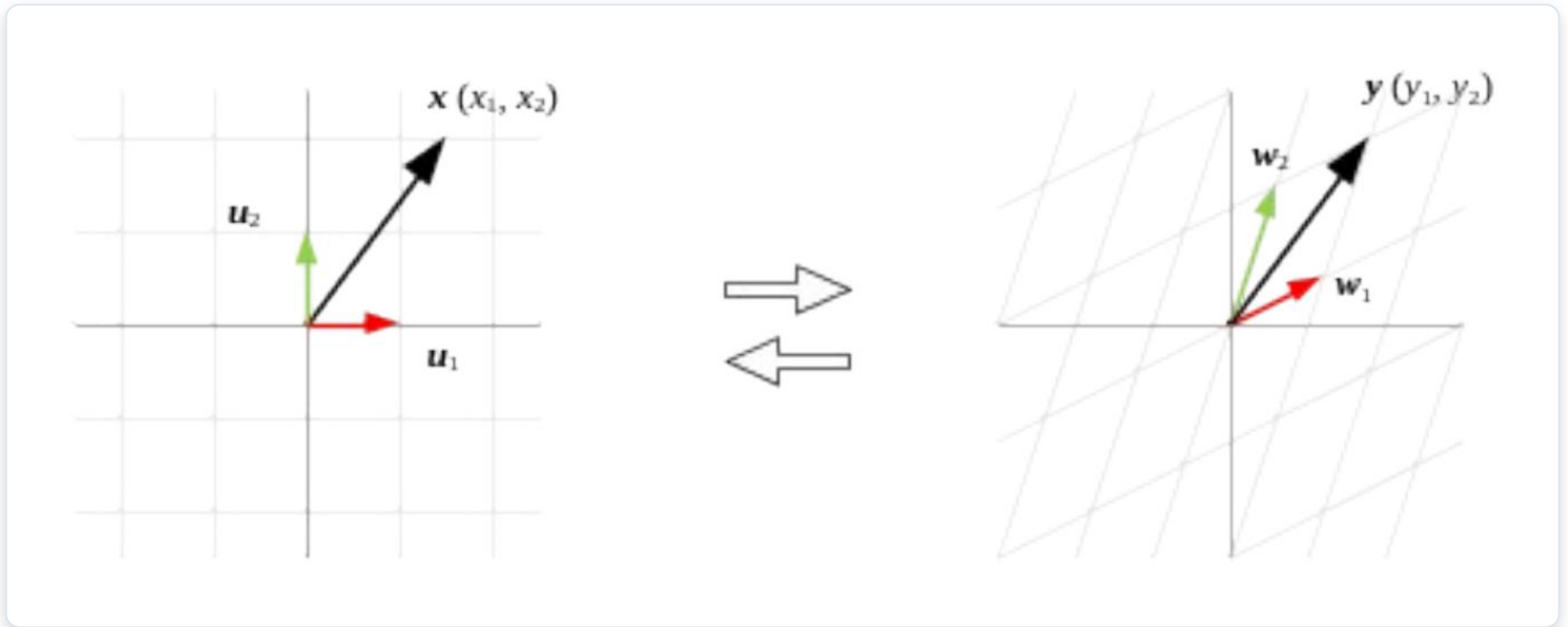
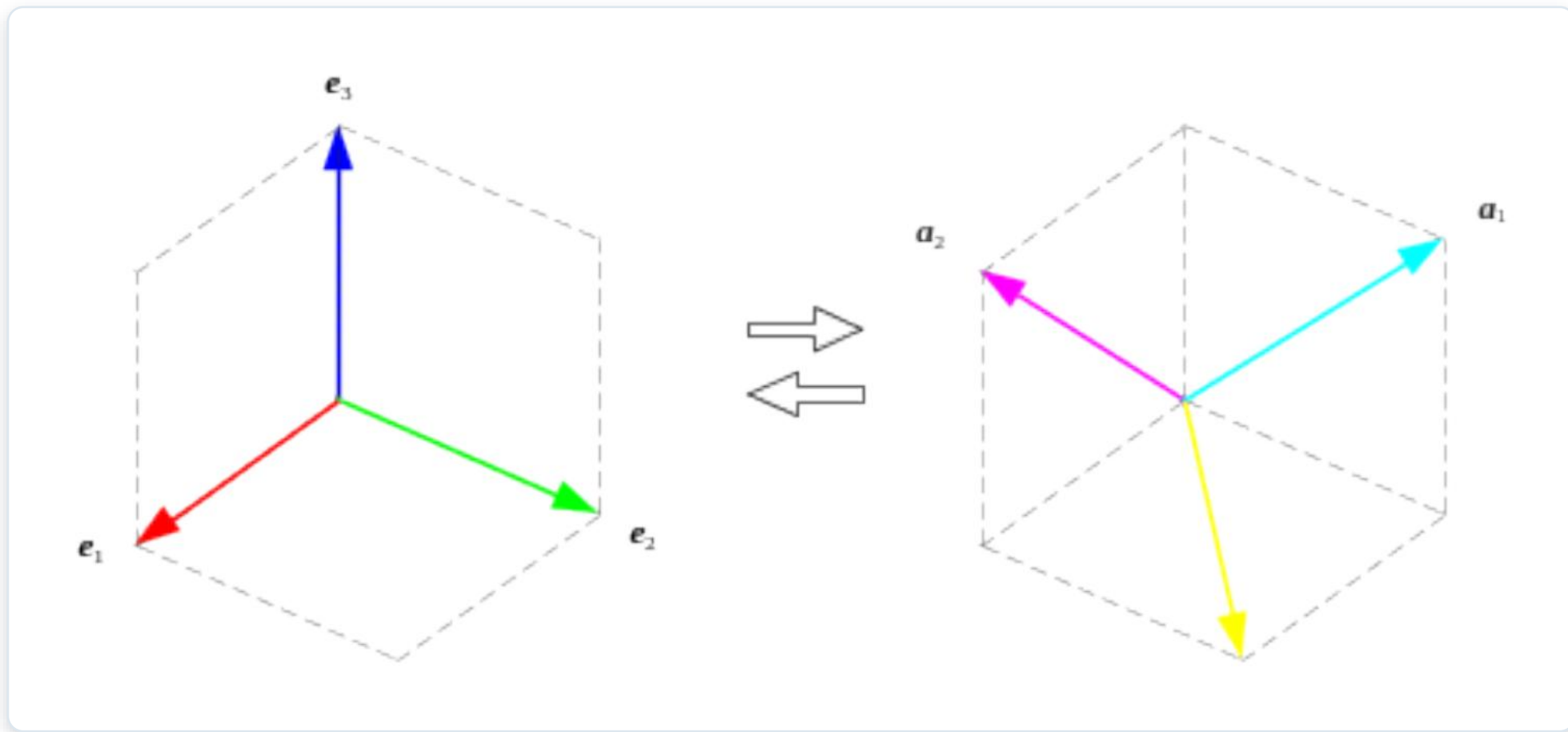


Figure 10 · Converting Between the Standard Basis and a Non-orthogonal Basis

- "Pure red" has coordinate $(-0.5, 0.5, 0.5)$ under the basis $\{a_1, a_2, a_3\}$



A Basis Is Never Unique: from Standard to Non-orthogonal

Standard Basis $\rightarrow$ Orthonormal Basis $\rightarrow$ Orthogonal Basis $\rightarrow$ Non-orthogonal Basis

- Constraints relax step by step, from "unit length + orthogonal" down to just "linearly independent"
- Change of basis $\mathbf{y} = \mathbf{W}^{-1}\mathbf{U}\mathbf{x}$ lets the same vector's coordinates convert freely between bases
- Converting between the RGB standard basis and the CMY non-orthogonal basis is a real application of change of basis

Glossary for This Section

Basis 基底 —— a linearly independent set of vectors that spans a vector space

Orthonormal Basis 规范正交基 —— a basis of pairwise orthogonal unit vectors

Non-orthogonal Basis 非正交基 —— basis vectors linearly independent but not necessarily orthogonal

Change of Basis 基底转换 —— converting the same vector's coordinates between bases

CMYK Color Model 基于非正交基 $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ 构造的印刷色彩空间

Standard Basis 标准正交基 —— the most natural orthonormal basis, e.g. $\{\mathbf{e}_1, \mathbf{e}_2\}$

Orthogonal Basis 正交基 —— basis vectors pairwise orthogonal, any length

Orthogonal Matrix 正交矩阵 —— a matrix $\mathbf{V}$ satisfying $\mathbf{V}\mathbf{V}^T = \mathbf{I}$

Inverse Matrix 逆矩阵 —— the key tool $\mathbf{A}^{-1}$ for finding coordinates under a new basis

Unit Vector 单位向量 —— a vector whose norm (L2 length) equals 1

Further Reading

Further Reading · Orthogonal Matrices and Rotation

Orthogonal matrices correspond to rotations and reflections — key to the geometric meaning of orthonormal bases

Further Reading · Applications of Change of Basis

Change of basis underlies PCA (principal component analysis), the Fourier transform, and many other methods

Review · Vector Spaces and Linear Combination

This section's basis concept builds on the vector space and `span()` ideas from §6.1 and §6.2

Open-Source Repository

Full code and image assets: <https://github.com/Visualize-ML>

Exercises Q1 – Q8

Q1 Use `numpy.linalg.matrix_rank()` to test whether the following vector pairs form a basis for the 2-D plane: $[1,0]$ & $[0,-1]$; $[1,0]$ & $[0,2]$; $[1,1]$ & $[2,2]$; $[0,0]$ & $[1,1]$; $[1,2]$ & $[2,1]$; $[3,4]$ & $[4,3]$.

Q3 Use `numpy.linalg.det()` to test whether the following vector triples form a basis for 3-D space: $[-1,0,0]$ & $[0,-1,0]$ & $[0,0,-1]$; $[1,1,1]$ & $[2,2,2]$ & $[3,3,3]$; $[1,1,0]$ & $[0,1,1]$ & $[1,0,1]$.

Q5 Use NumPy to compute red e_1 , green e_2 , and blue e_3 's coordinates under the CMY basis $[a_1, a_2, a_3] = [e_2 + e_3, e_1 + e_3, e_1 + e_2]$.

Q7 Write a function `is_basis(vectors)` that uses `numpy.linalg.matrix_rank()` to test whether a set of vectors can form a basis.

Q2 Use NumPy inner products to test which of the following vector pairs are orthogonal bases of the 2-D plane: $[1,0]$ & $[0,2]$; $[3,4]$ & $[4,-3]$; $[1,1]$ & $[-1,1]$; $[1,2]$ & $[3,1]$.

Q4 Use `numpy.linalg.inv()` to compute vector $a = [3,4]^T$'s coordinates under the bases $[1,0]$ & $[0,-1]$, $[1,0]$ & $[0,2]$, $[1,2]$ & $[2,1]$, and $[3,4]$ & $[4,3]$.

Q6 Use NumPy to verify that $\{e_1, e_2, e_3\}$ is the standard basis of 3-D space: pairwise inner products are 0, and every vector has unit norm.

Q8 Building on `is_basis`, write a function `is_orthogonal_basis(vectors)` that checks whether the vectors are pairwise orthogonal.

Exercises Q9 – Q16

Q9 Building on `is_orthogonal_basis`, write a function `is_orthonormal_basis(vectors)` that additionally checks each vector has unit norm.

Q11 Use Matplotlib to plot the unit-square grids for each of the four orthonormal bases in Figure 5.

Q13 Use `change_of_basis()` to verify: a vector with coordinate (3,4) under the standard basis, converted to a non-orthogonal basis `[a1,a2]` and back, returns to its original coordinate.

Q15 Randomly generate 100 invertible 2×2 matrices and count how many correspond to orthogonal bases (zero inner product between columns).

Q10 Construct an orthonormal basis rotated by 30° in NumPy and verify it satisfies $VV^T=I$.

Q12 Write a function `change_of_basis(v, U, W)` that converts vector `v`'s coordinate from basis `U` to basis `W`.

Q14 Use NumPy to verify that the two column vectors of any invertible 2×2 matrix form a basis for the plane.

Q16 Use SymPy symbolic computation to derive the general form of a 2-D rotation matrix $R(\theta)$ as an orthonormal basis, and verify $R(\theta)R(\theta)^T=I$.

Exercises Q17 – Q25

Q17 Use NumPy to verify that the CMY non-orthogonal basis $\{a_1, a_2, a_3\}$ has pairwise angles of 60° (hint: $\cos\theta = a \cdot b / (|a| |b|)$).

Q19 Use ipywidgets sliders to adjust the directions of a 2-D basis's two vectors, showing in real time how a fixed point's coordinates change under the new basis.

Q21 Use NumPy to verify that if matrix A has determinant 0, then A's column vectors cannot form a basis.

Q23 Use NumPy to verify that an orthogonal matrix's determinant is always +1 or -1.

Q25 Comprehensive exercise: use Python to render a 3-D visualization animation of the RGB→CMY change of basis, mimicking Figure 10.

Q18 Comprehensive exercise: write a function `rgb_to_cmy(color)` that uses change of basis to convert an RGB coordinate to a CMY coordinate.

Q20 Use a Matplotlib animation to show vector $a=[3,4]^T$'s coordinate point changing as the basis rotates gradually from the standard basis to some orthonormal basis.

Q22 Use `sklearn.decomposition.PCA` for a simple example explaining why choosing a good basis makes high-dimensional data easier to analyze.

Q24 Write a function `gram_schmidt(vectors)` implementing the Gram-Schmidt process to convert a linearly independent set into an orthogonal basis.

10 Prompts to Explore with AI

1 Explain why the standard basis is more "natural" than other orthonormal bases.

2 Prove that if matrix $\mathbf{V}$ satisfies $\mathbf{V}\mathbf{V}^T = \mathbf{I}$, then $\mathbf{V}$'s columns form an orthonormal basis.

3 Explain the containment relationship among orthogonal bases, orthonormal bases, and non-orthogonal bases, with one example of each.

4 Demonstrate the Gram-Schmidt process in Python, converting a linearly independent set of vectors into an orthogonal basis.

5 Explain why the CMY basis corresponding to the RGB standard basis is non-orthogonal, and what practical significance this has in printing.

10 Prompts to Explore with AI

6 Explain the geometric meaning of the change-of-basis formula $\mathbf{y} = \mathbf{W}^{-1}\mathbf{U}\mathbf{x}$, and why $\mathbf{U}\mathbf{x} = \mathbf{W}\mathbf{y}$ holds.

7 Give an example of change of basis in a real application, such as PCA, image compression, or the Fourier transform.

8 Explain why a vector space's basis is never unique, yet the number of vectors in any basis (the dimension) is always the same.

9 Use NumPy to verify that the same vector has different coordinates under different bases, yet remains the same geometric object.

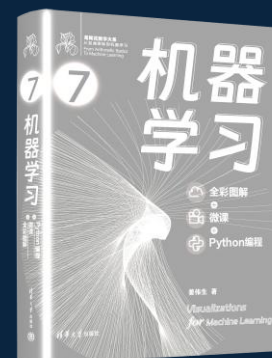
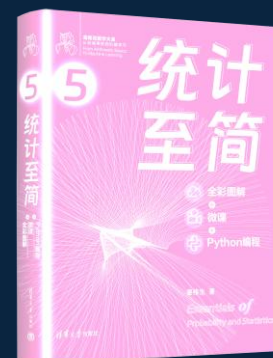
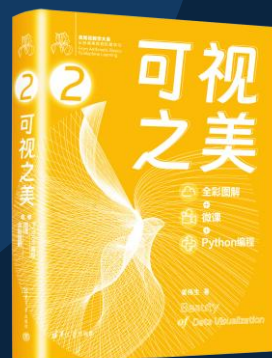
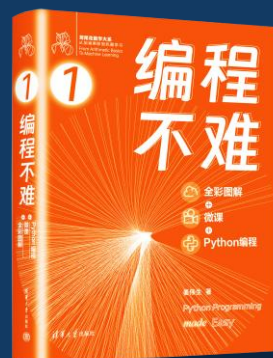
10 Design a Python exercise: given an arbitrary RGB color, compute its coordinate under the CMY non-orthogonal basis, and visualize the relationship between the two coordinate systems.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

